Lean Six Sigma Yellow Belt

Case Studies

Case Study 1: Reducing Inventory Waste in Precision Instrument Assembly

Background:
The company manufactures precision gas chromatographs for pharmaceutical labs. Due to complex part requirements, it maintains high inventory buffers to avoid stockouts.

Scenario:
A Lean Six Sigma team identified inventory holding as one of the key wastes (Muda), resulting in higher carrying costs and expired components.

Challenges:

- Overstocking sensitive components with short shelf lives.
- Inconsistent demand from clients with customised orders.
- Excessive time spent locating components on the shop floor.

Solution:
Implemented **5S** and **Just-in-Time (JIT)** principles. Used **Spaghetti Diagrams** to optimise part storage and reduce unnecessary motion. Integrated a pull system based on actual customer orders.

Outcome:
Inventory levels reduced by 35%, space freed up by 20%, and picking times dropped by 40%. Lead time to assemble a unit decreased by 2 days.

Case Study 2: Minimising Defects in Spectrometer Calibration

Background:
Calibration of spectrometers is crucial to product performance and must comply with strict ISO standards.

Scenario:
High rework rates were found during final product QA due to calibration errors detected late in the process.

Challenges:

- Manual data entry led to input errors.
- No standard procedure across calibration stations.
- Lack of root cause analysis for recurring calibration failures.

Solution:
Applied **DMAIC** methodology. Used **Cause & Effect (Fishbone Diagrams)** and **5 Whys** to identify root causes. Introduced **Poka-Yoke** mechanisms in software to avoid incorrect inputs.

Outcome:
Calibration error rates dropped by 60%, rework reduced by 50%, and compliance with ISO improved in internal audits.

Case Study 3: Improving Workflow Through Gemba Walks

Background:
An R&D line was introducing a new optical sensor prototype with several departments involved—design, engineering, QA, and procurement.

Scenario:
Miscommunication and workflow bottlenecks led to significant delays in prototype testing.

Challenges:

- Engineers and procurement unaware of each other's timelines.
- Missing parts at testing stage.
- Lack of visibility in the end-to-end process.

Solution:
Managers conducted **Gemba Walks** and **Process Stapling** to visualise real issues. Built a cross-functional **Value Stream Map**.

Outcome:
The team aligned their timelines, resulting in a 25% faster prototype development cycle and increased inter-departmental collaboration.

Case Study 4: Enhancing Process Capability for Cleanroom Assembly

Background:
High-sensitivity instruments were assembled in cleanrooms, but inconsistencies in output quality were affecting customer satisfaction.

Scenario:
Customer complaints rose due to particulate contamination detected post-delivery.

Challenges:

- Lack of SOP adherence among operators.
- Inconsistent process environment variables.
- No statistical monitoring in place.

Solution:
Introduced **SPC (Statistical Process Control)** and trained Yellow Belts to monitor parameters. Deployed **Standard Work** practices across all shifts.

Outcome:
Process capability improved from 1.2 to 1.8 (CpK), reducing complaints by 70% over two quarters and strengthening compliance with ISO 14644 standards.

Case Study 5: Shortening Lead Time for Custom Instrument Orders

Background:
Clients often requested bespoke analytical instruments with minor modifications to base models.

Scenario:
Despite small customisation, delivery timelines were long, impacting satisfaction and repeat business.

Challenges:

- Each order treated as a new product with complete cycle time.
- Repeated documentation efforts.
- Batch production mindset led to waiting.

Solution:
Used **Heijunka (Production Levelling)** and **SMED (Single-Minute Exchange of Die)** principles to shift from batch to flexible single-piece flow. Standardised 70% of documentation templates.

Outcome:
Average lead time for custom orders reduced by 30%, and customer repeat orders increased by 18% within 6 months.